

Statistical protocol: matched comparisons, intervals, and bounded claims I

No “robust beats naive” claim is written before a statistical test produces it (Algorithm Gate I). Where the active-inference community has typically reported belief-fusion outcomes as single illustrative runs, we treat every headline contrast as a paired hypothesis test with effect-size estimation, multiple-testing deflation, confidence intervals, and an observed-effect design-power calculation — the reporting discipline expected by robust statistics and applied inference (Huber and Ronchetti 2009; Efron and Tibshirani 1993; Koehler et al. 2009; Morris et al. 2019; Loy and Korobova 2021; Nakagawa and Cuthill 2007; Benjamini and Hochberg 1995; Wasserstein and Lazar 2016). The protocol lives in `statistics.py`, sits at the analysis tier above the locked FedGVI core, and emits every number the results sections report; nothing is hard-coded (ISC-30).

Paired comparison and standardized effect size I

The headline claim is *paired*: across matched scenarios — same seed, same contamination — does the robust aggregator raise consensus accuracy over the naive log-linear pool? For the expanded review grid, the inferential unit is the seed: trials are nested within a seed/cell and are averaged before seed-level contrasts. The primary robustness sweep also retains a trial-level paired diagnostic, but its matched trial replicates are nested within one fixed seeded world and are not independent worlds. The seed-level review-grid estimand is therefore the robust-minus-naive accuracy difference after the declared trial reduction, not a contrast of two independently drawn group means. Each configured robust method is retained in every review-grid rate panel and has its own seed-level contrast, interval, and test; no pooled-selected curve or pooled-selected inferential member enters that grid. The naive

Paired comparison and standardized effect size I

comparator is the project's log-linear pool eq. 6. Under the shared-support, posterior-log-potential, and fixed-weight assumptions of sec. 5.8, it specializes the message-combination term of Friston et al.'s Eq. 7 (2024), rather than reconstructing the complete source protocol. We test it with the matched-pairs Wilcoxon signed-rank test (Wilcoxon 1945) (`statistics.paired_test`, ISC-28), interpreted under the usual signed-rank conditions: seed-level paired differences are independent across the declared seed schedule, while the primary sweep's trial-level diagnostic remains conditional on its fixed world; the differences are ranked after zero differences are removed, and the null is a symmetric distribution of paired differences around zero rather than an equality-of-means claim (Fay and Proschan 2010). This is the right comparison for bounded, often non-Gaussian accuracy deltas, but it is

not assumption-free. The test reports the primary matched-pairs rank-biserial effect $r_{rb} = (T^+ - T^-)/(T^+ + T^-)$. We retain the monotone secondary transform $d_{eq} = 2r_{rb}/\sqrt{1 - r_{rb}^2}$, labeled a *rank-biserial-derived d-equivalent*, never raw Cohen's d and never a replacement for the mean contrast. When r_{rb} saturates at ± 1 , the transform diverges; tables print a signed saturation marker rather than a finite million-scale value. The primary-sweep headline display method RKL has rank-biserial effect 1.0000, d-equivalent saturated ($r=+1$) (large), with the paired mean difference and bootstrap interval reported alongside it.

Bootstrap interval estimates I

Every inferential mean — colony free energy, learning-curve KL, per-method accuracy, and the robust-minus-naive accuracy difference — carries a 95% percentile-bootstrap confidence interval (Efron and Tibshirani 1993) from `statistics.bootstrap_ci`, resampled from the declared unit. For the review grid, the unit is the seed after the nested trials have been reduced; for the primary fixed-world sweep, the interval is explicitly trial-level and conditional; for single-seed diagnostics, it is descriptive or trial-level. The single-colony mechanistic rate table is $n = 1$ per cell and is descriptive, not an inferential mean; its companion profile uses the declared nested design. This separation follows simulation-reporting guidance to declare the estimand and the independent Monte Carlo unit (Morris et al. 2019). The interval quantifies variation over the declared resampling unit (seed,

Bootstrap interval estimates I

recorded step, or matched trial) and is conditional on this simulation design, not on unmodeled real-world deployment uncertainty or alternative contamination models. The headline mean robust-minus-naive accuracy difference is 0.0846 with 95% CI [0.0821, 0.0873], and the accuracy at the verdict rate is 0.9021 for the naive pool (95% CI [0.8993, 0.9049]) against 0.9867 for the most accurate robust member (95% CI [0.9865, 0.9869]).

For every multi-seed summary we also report the Monte Carlo standard error ($\text{MCSE} = s / \sqrt{n_{\text{seed}}}$) and an approximate two-sided minimum detectable effect (MDE) at the configured target power. These are precision diagnostics over independent seeds, not claims about sampling a real population. The MDE uses a normal approximation conditional on the observed seed-level standard deviation (Koehler et al. 2009); it is reported alongside the non-parametric

Bootstrap interval estimates I

interval rather than used to replace the paired test.

Multiple-testing deflation by BH-FDR I

The sweep compares each robust divergence against the naive pool, so an uncorrected $p < 0.05$ across the family would manufacture false discoveries. We control the false-discovery rate with the Benjamini-Hochberg step-up procedure (Benjamini and Hochberg 1995) (`statistics.bh_fdr`, ISC-29) at $\alpha = 0.05$. The verdict family owns one robust-versus-naive contrast per robust method at the predeclared verdict rate; each rate table owns its own within-method rate family. Families are not pooled across figures or across the review-grid cells. The review-grid families retain all configured method contrasts and do not derive a selected member by pooled mean. The procedure returns both the rejection mask and the monotone BH q-values. BH controls expected false discovery proportion within the stated family, not the family-wise probability of any false positive; the

Multiple-testing deflation by BH-FDR I

manuscript therefore states the family each table uses. The positive-contrast rule is strict and conjunctive: a method is a BH-rejected positive contrast *iff* BH rejects its null **and** its rank-biserial effect is positive. This is a statistical decision for the named family, not a unique scientific winner. The primary-sweep headline display method has raw $p = 1.11 \times 10^{-158}$, deflating to BH $q = 1.11 \times 10^{-158}$.

Prospective power analysis for the verdict rate I

We report not only *that* the verdict rejected the null but the observed-effect planning power implied by the run and the number of pairs a confirmatory run could budget. The observed-effect design power of the headline paired Wilcoxon at the primary sweep's matched-trial unit, $n_{\text{trial}} = 960$, $\alpha = 0.05$, alternative greater, is 1.0000; the prospective sample size for the target power 0.80 at the observed effect is $n_{\text{trial}} = 5$. The estimator is the deterministic noncentral-normal approximation to the matched-pairs t -test, deflated by the Wilcoxon's Pitman asymptotic relative efficiency $3/\pi$, so the reported power is approximately calibrated for the signed-rank test the harness runs (the deflation is exact under a normal-shift alternative); note the power computation is directional for planning, while the reported p-values are two-sided.

Prospective power analysis for the verdict rate I

This is an observed-effect planning approximation conditional on the observed effect size, not independent evidence for the result and not a confirmatory power guarantee.

Reporting tables and the honesty boundary I

The results sections render five statistics tables straight from these tokens: the per-rate accuracy sweep (tbl. 3), the robust-versus-naive verdict (tbl. 5), the standardized-effect verdict with observed-effect design power and prospective n (tbl. 6), the per-method accuracy with bootstrap CIs at the verdict rate (tbl. 7), and the per-contamination-rate paired tests (tbl. 4). Every cell is a generated token, never hand-typed.

Reporting tables and the honesty boundary I

The honesty contract binds at exactly this tier. The effect-size, CI, and power enrichment above decorates the *server-side* robust_agg regate divergence-reweighting contrasts only. It does not certify the per-client β /rcce generalized-Bayes (FedGVI) guarantees, which are pinned by the locked core (Proposition 4 of sec. 6) rather than by these aggregation-level statistics — the three robustness axes of sec. 3.3 kept distinct.

Computational complexity and scaling diagnostic I

The release now reports computational complexity from the implementation itself. Let N be the number of agents, S the number of categorical states, I the solver-iteration budget, B the number of variational starts, and M the number of conditionally independent observation modalities. The dominant dense work and the storage actually retained by the current NumPy paths are:

Computational complexity and scaling diagnostic I

operation	dominant time order	retained or peak storage
log-linear pooling	$\Theta(NS)$	$\Theta(NS)$
iterative robust pooling	$\Theta(INS)$	$\Theta(NS + IS)$
objective-backed variational pooling	$\Theta(BINS)$	$\Theta(NS + IS + BS)$
self-excluding naive sharing	$\Theta(N^2S)$	$\Theta(NS)$
self-excluding robust sharing	$\Theta(IN^2S)$	$\Theta(NS + IS)$
one-step state inference	$\Theta(MS)$	$\Theta(S)$
server round, excluding queue/network wait	$\Theta(N \log N + INS)$	$\Theta(NS)$

Computational complexity and scaling diagnostic I

These are dominant interaction counts, not hardware-independent FLOP totals. The N^2 sharing term is material: with sensory attenuation enabled, one round computes one global pool and one leave-one-out pool per agent. The iterative server rules additionally retain their returned per-iteration histories, which is why their storage rows include IS . The server row includes worker-ID sorting and dense aggregation; incoming serialized belief volume is linear in NS , while queue and network latency are deliberately outside this local-compute accounting. The local path serializes one consensus-plus-weight result; if physical broadcast bytes are counted per recipient, that outgoing volume additionally has an N^2 term because each result carries the N agent weights.

Computational complexity and scaling diagnostic I

The accompanying seeded benchmark measures the real public call paths on the declared grids $N \in \{4, 8, 16, 32, 64\}$, $S \in \{256, 512, 1024, 2048, 4096\}$, self-excluding sharing $N \in \{4, 8, 16, 32\}$, and $M \in \{1, 2, 4, 8\}$. The fixed dimensions are $N = 256$, $S = 64$ for aggregation, and $S = 16384$ for state inference; direct aggregation and server timings use $I = 6$, while the public robust self-excluding sharing path uses its solver budget $I = 32$; the default variational path uses $B = 3$. Each grid point is warmed up 1 time(s) and measured 5 time(s), with median time plotted and the observed repeat range shown as a min-max bar. The fixed input seed is 20260728 on the *arm64* machine using Python 3.13.11 and NumPy 2.4.2.

Computational complexity and scaling diagnostic I

The measured log-log slopes are descriptive checks of the expected orders, not performance guarantees: agent-axis slopes are 0.88 (log-linear), 0.94 (iterative robust), 0.71 (variational), 1.59 (naive self-excluding sharing), and 1.94 (robust self-excluding sharing); state-axis slopes are 0.41, 0.41, and 0.42; the modality-axis inference slope is 0.67. The slope fit is a timing diagnostic on this machine, not an inferential test and not evidence that the same constants hold under another BLAS, accelerator, process topology, or distributed network. A finite grid can also yield a sublinear fitted slope when validation, allocation, cache, and interpreter overheads are material; the implementation-derived order is the governing claim, not equality between a finite-grid slope and its exponent.

Computational complexity and scaling diagnostic I

Figure fig. 1 visualizes the implementation-derived orders and the corresponding finite-grid timing diagnostic.

Computational complexity and scaling diagnostic I

Implementation complexity and machine scaling

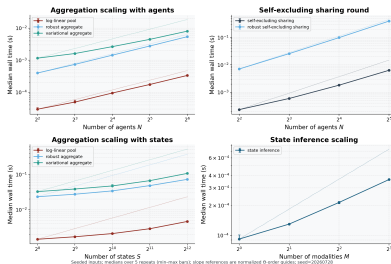


Figure 1: Implementation-derived complexity and seeded machine-scaling diagnostic. Source relation: original project computational-complexity diagnostic; estimand: median wall-clock time of the real categorical aggregation, naive and robust self-excluding sharing, and state-inference call paths as the declared dimension changes; uncertainty: min-max span over the repeated timings, not a confidence interval; replication unit: fixed seeded input at each grid point with the declared timing repeats. The x-axis is the varied agent, state, or modality dimension, and the y-axis is median wall-clock time in seconds. The panels show agent scaling for the aggregation rules, naive and iterative-robust N^2 leave-one-out sharing, state scaling for the aggregators, and modality scaling for one-step inference. Dotted lines are normalized Θ -order guides from the implementation-derived accounting; they are not fitted claims. The experiment ran on *arm64* with Python 3.13.11, NumPy 2.4.2, seed 20260728, and 5 measured repeat(s) after 1 warmup(s).